

Teaching Chemistry with Orange Data Mining

Pedagogical guidelines, classroom patterns, and ready-to-use examples for cheminformatics, QSAR, molecular standardization, ChEMBL curation, cyclic fingerprints, scaffolds, activity cliffs, and applicability domain.

Inquiry-first Students begin with a chemical question, not a button sequence.	Workflow literacy Orange workflows make data flow, assumptions, and model steps visible.
Interpretability Matched motifs, scaffolds, and AD outputs support chemical reasoning.	Reproducibility Every activity asks students to record data source, preprocessing, settings, and limits.

Design note

This booklet uses a clean modular style: short sections, visual cards, workflow boxes, checklists, and reusable lesson patterns. It is intended for print, PDF sharing, and teacher preparation.

Prepared for the OWChemInf / chem-inf-widgets teaching package

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How to use this booklet

Use it as a teacher-facing guide. The individual worksheet packages in docs/teaching/ provide student-facing step-by-step activities; this booklet explains how to teach with them.

1. Teaching philosophy for Orange in chemistry

Orange is valuable in chemistry teaching because it makes computational workflows visible. Students can see where molecular structures enter, where preprocessing happens, how descriptors are generated, how models are trained, and where interpretation or uncertainty enters the workflow.

Pedagogical goal

Do not teach Orange as a menu of widgets. Teach it as a way to express chemical reasoning as a transparent data workflow.

Three shifts in classroom practice

From calculations to decisions Students should ask whether a computational result is chemically meaningful, not only whether a widget produced a table.	From black boxes to evidence Each workflow should produce at least one interpretable output: matched motifs, scaffold summary, descriptor meaning, AD warning, or curation note.
From isolated tools to pipelines Real cheminformatics depends on linked steps: curation, standardization, descriptors, validation, interpretation, and reporting.	From correct answer to defensible method Students learn to justify preprocessing choices, split strategy, model selection, and limitations.

Recommended learning cycle

Step	Teacher prompt	Student output
Question	What chemical problem are we trying to answer?	A clear research or learning question.
Data	Where do the structures and activities come from?	Dataset with provenance notes.
Preparation	What must be cleaned or standardized?	Documented preprocessing choices.
Representation	Which descriptors or fingerprints are appropriate?	Justified feature representation.
Model / analysis	Which tool answers the question?	Model, visualization, search, or summary.

Interpretation	What chemical insight did we gain?	Evidence-based explanation.
Limits	Where should we not trust the result?	Applicability/domain/uncertainty statement.

TPACK, BLOOM, SAMR

2. Learning design principles

A strong Orange activity combines chemical content knowledge, pedagogical structure, and technological affordances. The tool should not replace chemistry; it should expose chemical assumptions and support reasoning.

TPACK alignment

Dimension	In chemistry teaching with Orange
Content knowledge	Molecular structures, activity data, descriptors, scaffolds, fingerprints, QSAR, AD, and chemical interpretation.
Pedagogical knowledge	Inquiry prompts, scaffolded worksheets, peer discussion, prediction-before-computation, reflection.
Technological knowledge	Orange widgets, RDKit-backed calculations, tables, workflows, visualization, exports.
Integrated TPACK	Students use computational workflows to answer chemical questions and justify methodological decisions.

Bloom progression

Bloom level	Example task
Remember	List common heterocycles or descriptor families.
Understand	Explain why standardization changes downstream fingerprints.
Apply	Build a workflow to compute descriptors and train a QSAR model.
Analyze	Compare random split and scaffold split performance.
Evaluate	Judge whether a prediction lies inside the applicability domain.
Create	Design a complete reproducible cheminformatics workflow for a new dataset.

SAMR progression

Substitution Use Orange instead of a spreadsheet formula to calculate descriptors.	Augmentation Automatically track matched motifs, failed structures, and output provenance.
Modification	Redefinition

Students compare multiple pipelines and interpret model reliability.

Students design a publishable, reproducible cheminformatics investigation.

REUSABLE SESSION DESIGN

3. Modern lesson architecture

The most effective sessions use a consistent lesson architecture. This reduces cognitive load and lets students focus on chemistry rather than software navigation.

The 75-minute practical session template

Time	Activity	Teacher role
0-10 min	Chemical problem framing and prediction-before-clicking.	Connect the workflow to prior chemistry knowledge.
10-20 min	Dataset inspection and quality discussion.	Ask students to identify possible data problems.
20-40 min	Guided workflow construction.	Demonstrate only the minimum needed steps.
40-55 min	Student exploration and parameter variation.	Circulate and ask justification questions.
55-70 min	Interpretation and comparison.	Focus on evidence, not just outputs.
70-75 min	Exit ticket.	Collect one insight, one limitation, one next question.

The worksheet page pattern

Context Short chemical scenario and why the task matters.	Learning outcomes Three to five measurable outcomes.
Workflow One minimal Orange workflow and one optional extension.	Tasks Stepwise work with prediction-before-computation.
Interpretation	Assessment

Guiding questions tied to chemistry.

Small rubric or checklist for evidence quality.

Workflow

File -> Mol Standardizer -> Descriptors / Fingerprints -> Model or Analysis -> Interpretation -> Report

FROM DATA TO CHEMICAL REASONING

4. Core workflow patterns

Data cleaning

Workflow: File -> Mol Standardizer -> Data Table

Use case: Use when students need to understand invalid SMILES, salts, charges, and reproducibility.

Fingerprint interpretation

Workflow: File -> Cyclic Registry Fingerprint -> Matched Registry Entries -> Data Table

Use case: Use when the goal is interpretable structural motifs.

QSAR baseline

Workflow: File -> Mol Descriptors 2 -> QSAR Regression -> Test & Score

Use case: Use for first QSAR modelling sessions.

Scaffold-aware validation

Workflow: File -> Mol Descriptors 2 -> Scaffold Splitter -> QSAR Regression

Use case: Use to show why random split can overestimate performance.

Applicability domain

Workflow: Reference File -> Query File -> Applicability Domain -> Data Table

Use case: Use to teach prediction reliability.

Curation pipeline

Workflow: ChEMBL Retrieval -> Mol Standardizer -> Duplicate Aggregation -> QSAR-ready Table
Use case: Use to teach real-world bioactivity data limitations.

WHAT IS ALREADY INCLUDED

5. Teaching package map

The package now contains multiple teaching modules. Use the map below to select a module based on learning goals.

Teaching package	Best for	Typical output
Cyclic Registry Fingerprint	Heterocycles, interpretable fingerprints, motif matching.	4096-bit table and matched registry entries.
QSAR Widgets	Descriptor preparation, fingerprint QSAR, model comparison.	Model performance and feature interpretation.
Applicability Domain	Reliability of predictions and external query sets.	Inside/outside-domain labels and AD summary.
Molecular Standardization	Data cleaning and reproducible preprocessing.	Standardized structures and standardization report.
ChEMBL Bioactivity Curation	Real data retrieval and QSAR-ready dataset preparation.	Curated activity table with provenance.
Scaffold and Activity Cliff Analysis	Medicinal chemistry series, validation, cliffs.	Scaffold summaries, splits, cliff pairs.
Supplementary Cheminformatics Workflows	Similarity, SMARTS, diversity, filters, MMP, reactions.	Library design and QC outputs.

FAST LESSON IDEAS

6. Ten classroom-ready examples

Example	Student task	Main widgets
1. Why preprocessing matters	Students compare fingerprints before and after molecular standardization.	Mol Standardizer + Fingerprint Generator
2. Heterocycles in drugs	Students identify matched cyclic registry entries in drug-like molecules.	Cyclic Registry Fingerprint
3. Aromatic vs Kekule forms	Students test whether different SMILES representations trigger similar motif matches.	Cyclic Registry Fingerprint
4. Descriptor meaning	Students connect logP, TPSA, HBD/HBA, and molecular weight with chemical intuition.	Mol Descriptors 2
5. First QSAR model	Students build a simple regression model and evaluate baseline performance.	QSAR Regression

6. Scaffold split challenge	Students compare random validation with scaffold-aware validation.	Scaffold Splitter
7. Activity cliffs	Students find similar molecules with large activity differences.	Activity Cliff Finder
8. Prediction reliability	Students decide whether external predictions are inside the applicability domain.	Applicability Domain
9. ChEMBL curation	Students filter bioactivity data and document curation decisions.	ChEMBL widgets + Mol Standardizer
10. Library design	Students combine filters, diversity picking, and motif analysis to select compounds.	Drug Filter + Diversity Picker + Cyclic Registry

EVALUATE REASONING, NOT CLICKING

7. Assessment and rubrics

Assessment should reward chemical reasoning, methodological transparency, and awareness of limitations. A visually correct workflow is not sufficient if students cannot explain why each step is used.

General 15-point rubric

Criterion	0	1	2	3
Data provenance	Absent	Source named only	Source and key filters listed	Complete source, filters, version/date, and limitations
Workflow construction	Missing	Partly incorrect	Correct basic workflow	Correct workflow with justified optional steps
Chemical interpretation	Absent	Superficial	Mostly correct	Specific, evidence-based, chemically sound
Validation / reliability	Absent	Metric reported only	Metric interpreted	Metric plus AD/split limitations discussed
Reproducibility	Absent	Some settings	Most settings	Full settings, outputs, and failure handling

Exit ticket prompts

- Which preprocessing choice most influenced the result?
- Which output is chemically interpretable and why?
- Where would you not trust the model or analysis?
- What would you change before using this workflow for a publication?

8. Accessibility, inclusion, and classroom management

Reduce cognitive load Provide a minimal starter workflow before asking students to explore variants.	Use paired roles One student drives the workflow; the other records assumptions and interpretation.
Make outputs visible Ask students to screenshot or export tables that support their conclusions.	Normalize errors Invalid molecules and failed calculations are part of real cheminformatics, not student failure.

Inclusive design checklist

- **Before class:** Test the environment, provide a small backup dataset, and prepare a screenshot of the expected workflow.
- **During class:** Pause after each major widget and ask students to predict the output before running it.
- **For mixed skill levels:** Provide optional extension tasks rather than accelerating the entire class.
- **For assessment:** Grade explanations and evidence more than speed.

COMMON ISSUES

9. Troubleshooting and reproducibility

Problem	Likely cause	Teaching response
Widget not visible	Package not installed in the active environment.	Run <code>pip install -e .</code> and restart Orange.
No matched motifs	Wrong SMILES column, invalid structures, or motifs absent.	Inspect input and compare with a known positive control.
Model performance too high	Data leakage, duplicate compounds, or random split optimism.	Use scaffold split and check duplicates.
External predictions questionable	Query compounds outside training domain.	Use applicability domain and report limitations.
Different results after standardization	Charges, salts, aromaticity, or tautomeric changes.	Compare original and standardized SMILES.

Minimum reproducibility record

Record this for every workflow

Dataset name and source; date/version; SMILES column; standardization profile; descriptor/fingerprint settings; split method; model settings; validation metric; AD method if used; exported outputs; known failures.

FROM ONE LAB TO A CURRICULUM

10. Implementation roadmap

The teaching materials can be used as individual exercises or assembled into a short cheminformatics module. The roadmap below supports gradual adoption.

Week	Focus	Suggested package
1	Molecular data and standardization	Molecular Standardization
2	Descriptors, fingerprints, and motif interpretation	Cyclic Registry Fingerprint + QSAR Widgets
3	Model building and validation	QSAR Widgets + Scaffold Splitter
4	Reliability and applicability domain	Applicability Domain
5	Real data curation	ChEMBL Bioactivity Curation
6	Medicinal chemistry interpretation and capstone	Scaffold and Activity Cliff + Supplementary Workflows

Capstone deliverable

Students produce a short reproducible report: chemical question, dataset source, curation decisions, standardized structures, descriptors/fingerprints, validation strategy, model or analysis results, applicability-domain discussion, and chemical interpretation.

Final recommendation

Use Orange to make the workflow visible, but always bring the discussion back to chemistry: structure, data quality, assumptions, uncertainty, and chemical meaning.

Appendix. Visual and editorial style guide

Use this compact style guide to keep future worksheets and booklets visually consistent.

Element	Guideline
Tone	Clear, student-facing, evidence-oriented, with short paragraphs and explicit outputs.
Color system	Navy for structure, teal for guidance, orange for activities, red for warnings, green for successful interpretation.
Worksheet design	Start with context and ILOs; include one minimal workflow, one extension, guiding questions, and a rubric.
Tables	Use compact rows and avoid overloading a page with too many columns.
Workflow text	Use File -> Standardizer -> Descriptor -> Model notation for accessibility and easy copying.
Assessment	Prefer rubrics that reward interpretation and reproducibility, not only software output.